ABOUT HERDER ELEKTRONISCHE SYSTEMEN

Engineering systems that can be understood, *tested and trusted.*

Herder Elektronische Systemen is an independent engineering practice focused on embedded controls, power electronics, real-time testing, technical recovery and AI-assisted engineering software.

01 / ORIGINS

It began with a personal *power-electronics laboratory.*

Herder Elektronische Systemen did not begin as a general consulting brand. It grew out of a personal engineering laboratory built by Jonathan Folkema to work directly with power electronics, embedded control, electric machines, instrumentation and real-time testing.

The laboratory started with development boards, test equipment, power stages, motors, sensors and experimental circuits. Over time, those individual tools became a working environment for developing complete systems: hardware connected to firmware, firmware connected to controls, controls connected to instrumentation, and measurements connected back to design decisions.

That environment shaped the way the firm works today. Problems are not treated as isolated software, electrical or mechanical tasks. They are followed across the entire technical chain until the physical behavior, implementation and evidence agree.

As the projects grew, another difficulty became impossible to ignore. Engineering work was often scattered across source files, models, test data, schematics, notes, archived revisions and incomplete documentation. Recovering the reasoning behind a project could become as difficult as building the project itself.

That problem led to work in technical recovery, evidence preservation and AI-assisted engineering software. SafeFileAI and SafeECADAI emerged from the need to keep engineering conclusions connected to the files, measurements and decisions that support them.

Herder Elektronische Systemen grew from that sequence of work: build the system, measure what it does, preserve the evidence and leave the project in a condition that another engineer can understand and continue.

THE LABORATORY FOUNDATION

- Power-electronics prototyping
- Embedded control hardware
- Electric-machine test systems
- PCB development and bring-up
- Oscilloscopes, electronic loads and instrumentation
- Real-time control and automated data collection
- Hardware, firmware and model integration

02 / TECHNICAL INDEX

A capability profile, *not a résumé.*

The technical background behind the work spans implementation, physical integration, measurement and documentation.

EMBEDDED

Embedded C/C++ and firmware

CONTROLS

Motor control and real-time systems

COMMUNICATIONS

CAN, UART, SPI and I^2C

MODELING

MATLAB and Simulink workflows

HARDWARE

PCB design, bring-up and validation

POWER

Power-converter prototyping

TESTING

Automated test and instrumentation

RECOVERY

Repository recovery and technical documentation

SOFTWARE

AI-assisted engineering tooling

03 / ENGINEERING PRINCIPLES

Evidence keeps the *system honest.*

01

Evidence before claims

Describe what the hardware, test or record supports.

Ø2

Preserve system context

Keep interfaces, assumptions and constraints attached.

Ø3

Make changes traceable

Connect implementation changes to the decision they serve.

Ø4

Test the real integration

Exercise the path between code, hardware and measurement.

Ø5

Document uncertainty

State what remains unknown and what would resolve it.

Ø4 / GOOD-FIT ENGAGEMENTS

Technical work with *enough substance.*

The strongest fit includes:

- Embedded or electromechanical prototypes

- Incomplete or inherited engineering projects
- Firmware and hardware integration
- Test automation and instrumentation
- Design recovery and technical documentation
- Technically demanding early-stage product development

05 / FOUNDER AND ENGINEERING LEAD

Jonathan Folkema

Jonathan Folkema is a mechanical engineer whose work developed through years of independently building laboratory capability around power electronics, embedded systems, electric machines, control systems and technical instrumentation.

His experience includes permanent-magnet motor drives, field-oriented control, CAN-connected embedded platforms, real-time test systems, precision motion equipment, PCB design and bring-up, converter prototyping, automated data collection and recovery of complex engineering repositories.

The distinguishing feature of his work is not any one discipline. It is the ability to move between them.

A control problem may require inspecting the power stage. A firmware problem may originate in sensing or timing. A weak test result may come from instrumentation rather than the design itself. An apparently complete project may still be unusable because its files, assumptions and evidence are disconnected.

Jonathan founded Herder Elektronische Systemen to address those interfaces directly. The practice is built around hands-on investigation, measured validation, careful recovery and documentation that preserves what is known, what remains uncertain and what should happen next.

WORKING PERSPECTIVE

Build enough of the system to understand the interfaces.

Measure the physical behavior before making strong claims.

Keep software conclusions connected to engineering evidence.

Preserve uncertainty instead of concealing it.

Leave projects easier to inspect and continue than they were found.

START A TECHNICAL DISCUSSION

Start with the *technical question.*

Describe the system, its current stage, the problem you are trying to resolve and what evidence is available. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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